



Oregon Institute of Technology

Computer Systems Engineering Technology Department

CST 407 – Python for AI

Homework 4

In this homework, you will be using the Reuters dataset. You will classify news related to 46 classes. While uploading data, make sure you limit your vocabulary to the **top 10,000 most frequent words**.

1. Load Reuters data using the command given below (make sure you limit your vocabulary to the **top 10,000 most frequent words**)
`(x_train, y_train), (x_test, y_test) = reuters.load_data(num_words=max_words)`
2. Pad the training and testing data points to ensure they have the same size of 200
3. Perform the necessary data manipulations to make sure they are ready to feed into NNs.
4. Use an Embedding layer and Dense layer. Choose appropriate activation functions, optimizer, loss function, and metrics. Run your NN model with 25% validation.
5. Use an Embedding layer and SimpleRNN layer. Choose appropriate activation functions, optimizer, loss function, and metrics. Run your NN model with 25% validation.
6. Use an Embedding layer and LSTM layer. Choose appropriate activation functions, optimizer, loss function, and metrics. Run your NN model with 25% validation.
7. Use an Embedding layer and 1D CNN layer. Choose appropriate activation functions, optimizer, loss function, and metrics. Run your NN model with 25% validation.
8. Now we want to use **glove.6B.100d.txt** from **glove6B** pretrained coefficients folder. (Link for glove6B is posted in the announcements section of the Lecture Canvas)
9. Create the embedding matrix using the floating format coefficients present in **glove.6B.100d.txt**.
10. Now repeat steps from 4 to 7. Make sure the Embedding layer is using the embedding matrix created in step 9.
11. Submit your Jupyter notebook (make sure every step is present in it). You **do not** need to submit a separate report.